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Terminalx

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What are Wheels?

Installing Wheels

Before introducing wheels, let's create a virtual environment.

```
mkdir wheels
cd wheels
python3 -m venv wheels-env
source wheels-env/bin/activate
```

Before we begin working with wheels, we need to make sure our system has the latest tools we need. Remember, the `-U` flag means to upgrade the package.

```
python -m pip install -U pip wheel setuptools
```

Installing a package often means downloading a wheel. You probably never noticed this as most people don't read the terminal output during the installation process. Take a look at what happens when we install the `numpy` package.

```
python3 -m pip install numpy
```

Python downloads a file that ends with the `.whl` file extension. That is a wheel. We can break this process down into two steps. One, download the wheel. Two, install the wheel. This whole process does not take much time at all.

```
Collecting numpy
  Downloading numpy-1.19.5-cp36-cp36m-manylinux2010_x86_64.whl (14.8 MB)
    |#####| 14.8 MB 10.0 MB/s
Installing collected packages: numpy
Successfully installed numpy-1.19.5
```

Compare this to installing a package that downloads the source distribution instead of a wheel. Install version 2 of the `uwsgi` package. Pay particular attention to the lines of output in the terminal.

```
python -m pip install 'uwsgi==2.0.*'
```

Python downloads a `tar.gz` file. This is a zipped version of the source distribution of the package. After that has been downloaded, Python says that it is building a wheel. Once Python built the wheel, the actual installation process begins. This is a noticeably longer process as there are more steps. Plus, building a wheel takes time.

```
Collecting uwsgi==2.0.*
  Using cached uwsgi-2.0.20.tar.gz (804 kB)
  Preparing metadata (setup.py) ... done
Building wheels for collected packages: uwsgi
  Building wheel for uwsgi (setup.py) ... done
  Created wheel for uwsgi: filename=uWSGI-2.0.20-cp36-cp36m-linux_x86_64.whl size=5
  Stored in directory: /home/codio/.cache/pip/wheels/4a/60/83/bd5b22ba1a9298cc00c9f
Successfully built uwsgi
Installing collected packages: uwsgi
Successfully installed uwsgi-2.0.20
```

Wheels and Source Distribution

So, what is a wheel and what is source distribution? A source distribution is the necessary source files and associated metadata that `pip` needs to install a package. In contrast, a wheel is a built distribution. That is, the source distribution is built into a form that is ready to be installed on your system.

Regardless of how you install a package, the process goes from source distribution, to built distribution, to installation on your system. Using wheels means the build step is done before you download the package. That is why the installation process with wheels is faster than source distribution. In addition, wheels are smaller than the source distribution, so the initial download should be quicker as well.

► Why are they called wheels?

Reading Question

Select all of the benefits of using a wheel. **Hint**, there is more than one correct answer.

☐ Smaller file

☐ Already built

☐ Packages run faster

☐ Downloads faster

The correct answers are:

- Smaller file
- Downloads faster
- Already built

Wheels provide all of the above advantages. However, using a wheel over source distribution does not affect how fast the Python code runs once the package has been installed.

Check It!

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